

Siphelele Dlamini

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PROFESSIONAL PROFILE

Highschool-taught programmer turned Machine Learning enthusiast with 6 months of professional experience building scalable web applications. Proven ability to apply design patterns for maintainability, tutored tertiary students, and automated workflows across projects.

I have theoretical and practical appreciation of various aspects of technology, with a clear focus on how it can be applied in resolving operational inefficiencies. Additionally, with a creative ability to think enthusiastic about my self development, I am currently completing research project; Financial Fraud Detection in Mobile Money Systems Using Supervised Machine Learning.

EDUCATION

University of Pretoria **2022 – 2025**
Bachelor of Science in Computer Science *GPA: 65 (BSc)*

- **Coursework:** AI, Software Eng & Modelling, Computer Networks/Security/Architecture, DB Systems, DSA, Calculus, Statistics, Discrete structures, Compiler Construction.

University of Pretoria **2025 – 2026**
Bachelor of Sciences with Honours in Computer Science *GPA: 74 (BScHons)*

- **Coursework:** Security & Machine Learning, with advanced coursework in Natural Language Processing, Data Mining, Digital Forensics & Investigation, Cybersecurity and Software Eng.

RESEARCH

In Progress

“Financial Fraud Detection in Mobile Money Networks Using Supervised Machine Learning”

COS700 Honours Research, University of Pretoria

- Analyzing mobile money transaction logs using a synthetic financial dataset generated via PaySim to engineer a Machine Learning (ML) detection prototype.
- Evaluating whether cost-sensitive learning (CSL) improves fraud detection performance on highly imbalanced mobile money datasets by incorporating the unequal costs of false positives and false negatives.
- Dynamically prioritizing the asymmetric business costs of misclassifications, balancing the financial loss of fraudulent transactions against the operational damage of falsely freezing legitimate accounts.

EXPERIENCE

Free-Lance AI-Assisted Developer **March 2026 – Current**
Self-Employed *Remote*

- Software consultancy specializing in application development, working on Law-Firm-System and Risk-Advisory-System.
- Embedded cutting-edge AI coding tools, agents, and LLMs directly into SDLC to accelerate client delivery times while maintaining architectural integrity and code quality.
- Utilized Replit, Claude, LangChain, and LangGraph for multi-agent workflows, and GitHub Copilot for CI/CD deployments.

Internship Software Developer **October 2024 – March 2025**
Intalos AI (startup)

- Contributed to the development of an AI-driven chatbot builder using Python and Django to automate customer interactions.

- Implemented a bot error detection system that manages and sends critical log data to the development team when a client chatbot crashes.
- Developed a universal LLM feature allowing clients to set the context, retrieval, and evidence of conversations to limit hallucination.
- Gained experience with AWS Serverless technologies, DevOps, Git, and full-stack solutions processing large-scale data.

Teaching Assistant

University of Pretoria

Feb 2023 – Nov 2026

Pretoria, South Africa

- Served as a Teaching Assistant across multiple semesters for Database Systems, Data Structures & Algorithms, Computer Networks, Imperative Programming, and Intro to Computer Science.

PROJECTS

BMW Automotive IT Colloquium 2026 (Dccy)

Tools: ReactJS, NodeJS, LangChain/LangGraph

- Designed, implemented, and presented a small-scale POC for an AI-native application to optimize the onboarding process at the ZA Hub, utilizing a hub-and-spoke multi-agent architecture alongside a five-member team.
- Automated the setup and orientation process for new engineers, autonomously handling Jira tickets and qualifying relative Confluence documentation.

FinGround

[Repo Link](#)

Tools: ASP.NET Web API, Postgres, Postman

- Designed an API using the CQRS pattern & Clean Architecture simulating a banking environment for third-party backend transactions and account monitoring.
- The Sandbox consists of features such as rate limiting for DDoS prevention, OAuth2 with mTLS to secure every request... etc

Workwise

[Repo Link](#)

Tools: VueJS, SocketIO, NodeJS, NestJS, MongoDB, Docker, Apache

- Developed a PWA for the service delivery industry (electrical contractors & plumbing companies) as a Software Eng Capstone Project.
- Streamlined operations with features like real-time inventory tracking and notifications.

SKILLS & COMPETENCIES

- **Languages:** JavaScript, TypeScript, PHP, NodeJS, Python (Data mining & Web development), C++, C#, Java.
- **Frameworks & Tools:** React, Vue, Django, ASP.NET10 API, PyTorch, PostgreSQL, MongoDB, Docker, Git.
- **Applications:** Microsoft Word, Excel, PowerPoint, Visio, LaTeX, Power Automate.
- **Competencies:** Problem-solving, Adaptability, Meta-Cognitive Thinking, Project-Based Worker.

INTERESTS

AI Security (GRC), RegTec, MoMo & Prediction Markets, cross-country, gaming